

Problem 18.13

Julie, Chris, and Allen will share an annual perpetuity-immediate of \$1,200. Julie will receive the first 9 payments. Chris will receive the next 16 payments. Allen will receive all remaining payments. At an annual effective interest rate of 5%, order the value of each person's share of the perpetuity.

Solution

The effective annual interest rate is

$$i = 0.05,$$

so

$$v = (1.05)^{-1}.$$

Julie

Julie receives the first 9 payments, so her present value is

$$1200a_{\overline{9}|}.$$

Thus,

$$\begin{aligned} J &= 1200 \left(\frac{1 - (1.05)^{-9}}{0.05} \right) \\ &\approx 8529.3860. \end{aligned}$$

Chris

Chris receives the next 16 payments, beginning after Julie's 9 payments. His present value is

$$1200a_{\overline{16}|}v^9.$$

Thus,

$$\begin{aligned} C &= 1200 \left(\frac{1 - (1.05)^{-16}}{0.05} \right) (1.05)^{-9} \\ &\approx 8383.3475. \end{aligned}$$

Allen

Allen receives all remaining payments after the first 25 payments. His share is a deferred perpetuity-immediate, so the present value is

$$1200a_{\infty}|v^{25}.$$

Thus,

$$\begin{aligned} A &= 1200 \left(\frac{1}{0.05} \right) (1.05)^{-25} \\ &\approx 7087.2665. \end{aligned}$$

Therefore,

$$J > C > A.$$

Hence, the order of the values is

$$\boxed{\text{Julie} > \text{Chris} > \text{Allen}}.$$